

Multimodal Autonomous Agent Applications

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1. Concept Introduction

1.1 What is an "Autonomous Agent"?

In the `largetmodel` project, the **multimodal autonomous agent** is the highest level of intelligence. It no longer simply responds to a user's command once, but is able to **autonomously think, plan, and continuously invoke multiple tools to complete a task to achieve a complex goal.**

The core of this functionality is the `agent_call` tool or its underlying **toolchain manager** (`ToolChainManager`). The autonomous agent is activated when a user makes a complex request that cannot be completed with a single tool call.

1.2 Implementation Principles

The autonomous agent implementation in `largetmodel` follows the industry-leading **ReAct (Reason + Act)** paradigm. Its core idea is to mimic the human problem-solving process, cycling between "thinking" and "acting".

1. **Reason:** When the agent receives a complex goal, it first invokes a powerful Language Model (LLM) to "think." It asks itself, "What should my first step be to achieve this goal? Which tool should I use?" The LLM's output is not a final answer, but an action plan.
2. **Act:** Based on the LLM's reasoning, the agent executes the corresponding action—calling `ToolsManager` to run the specified tool (such as `visual_positioning`).
3. **Observe:** The agent retrieves the result of the previous action ("observation"), for example, `{"result": "The cup was found, located at [120, 300, 180, 360]}"`.
4. **Rethink:** The agent submits the observation results, along with the original goal, back to the LLM for a second round of "reasoning." It asks itself, "I've found the cup's location. What should I do next to find out its color?" The LLM might generate a new action plan, such as `{"thought": "I need to analyze the image of the area where the cup is located to determine its color", "action": "seewhat", "args": {"crop_area": [120, 300, 180, 360]}}`.

This **think -> act -> observe** loop continues until the initial goal is achieved, at which point the agent generates and outputs the final answer.

2. Project Architecture

Key Code

1. Agent Core Workflow (`largemodel/utils/ai_agent.py`)

The `_execute_agent_workflow` function is the Agent's main execution loop, defining the core "planning -> execution" process.

```
# From largemodel/utils/ai_agent.py

class AIAgent:
    # ...

    def _execute_agent_workflow(self, task_description: str) -> Dict[str, Any]:
        """
        Executes the agent workflow: Plan -> Execute.
        """
        try:
            # Step 1: Mission Planning
            self.node.get_logger().info("AI Agent starting task planning phase")
            plan_result = self._plan_task(task_description)

            # ... (Return early if planning fails)

            self.task_steps = plan_result["steps"]

            # Step 2: Follow all steps in order
            execution_results = []
            tool_outputs = []

            for i, step in enumerate(self.task_steps):
                # 2.1. Process data references in parameters before execution
                processed_parameters =
                self._process_step_parameters(step.get("parameters", {}), tool_outputs)
                step["parameters"] = processed_parameters

                # 2.2. Execute a single step
                step_result = self._execute_step(step, tool_outputs)
                execution_results.append(step_result)

                # 2.3. If the step succeeds, save its output for reference in
                subsequent steps
                if step_result.get("success") and
                step_result.get("tool_output"):
                    tool_outputs.append(step_result["tool_output"])
                else:
                    # If any step fails, abort the entire task
                    return { "success": False, "message": f"Task terminated
                    because step '{step['description']}' failed." }

            # ... Summarize and return the final result
            summary = self._summarize_execution(task_description,
            execution_results)
```

```
        return { "success": True, "message": summary, "results":
execution_results }
```

```
# ... (Exception handling)
```

2. Mission planning and LLM interaction (`largemodel/utils/ai_agent.py`)

The core of the `_plan_task` function is to build a sophisticated prompt and use the reasoning ability of the large model to generate a structured execution plan.

```
# From largemodel/utils/ai_agent.py
```

```
class AIAgent:
```

```
# ...
```

```
def _plan_task(self, task_description: str) -> Dict[str, Any]:
    """
```

```
    Uses the large model for task planning and decomposition.
    """
```

```
    # Dynamically generate a list of available tools and their descriptions
```

```
    tool_descriptions = []
```

```
    for name, adapter in
```

```
self.tools_manager.tool_chain_manager.tools.items():
```

```
        # ... (Get the tool description from adapter.input_schema)
```

```
        tool_descriptions.append(f"- {name}({params}): {description}")
```

```
    available_tools_str = "\\n".join(tool_descriptions)
```

```
    # Build a highly structured plan
```

```
    planning_prompt = f"""
```

```
As a professional task planning agent, please break down user tasks into a series
of specific, executable JSON steps.
```

```
*** Available Tools:**
```

```
{available_tools_str}
```

```
*** Core Rules:**
```

```
1. **Data Passing**: When a subsequent step requires the output of a previous
step, it must be referenced using the `{{{steps.N.outputs.KEY}}}` format.
```

```
    - `N` is the step ID (starting at 1).
```

```
    - `KEY` is the specific field name in the output data of the previous step.
```

```
2. **JSON Format**: Must strictly return a JSON object.
```

```
*** User Tasks:**
```

```
{task_description}
```

```
"""
```

```
    # Calling large models for planning
```

```
    messages_to_use = [{"role": "user", "content": planning_prompt}]
```

```
    # Note that the general text reasoning interface is called here
```

```
    result = self.node.model_client.infer_with_text("",
```

```
message=messages_to_use)
```

```
    # ... (parsing the JSON response and returns a list of steps)
```

3. Parameter processing and data flow implementation (`largemodel/utils/ai_agent.py`)

The `_process_step_parameters` function is responsible for parsing placeholders and implementing data flow between steps.

```
# From largemodel/utils/ai_agent.py

class AIAgent:
    # ...
    def _process_step_parameters(self, parameters: Dict[str, Any],
                                previous_outputs: List[Any]) -> Dict[str, Any]:
        """
        Parses parameter dictionary, finds and replaces all {{...}} references.
        """
        processed_params = parameters.copy()
        # 正则表达式用于匹配 {{steps.N.outputs.KEY}} 格式的占位符
        pattern = re.compile(r"\\{\\{steps\\. (\\d+)\\.outputs\\. (.*?)\\}\\}")

        for key, value in processed_params.items():
            if isinstance(value, str) and pattern.search(value):
                # 使用 re.sub 和一个替换函数来处理所有找到的占位符
                # 替换函数会从 previous_outputs 列表中查找并返回值
                processed_params[key] = pattern.sub(replacer_function, value)

        return processed_params
```

Code Analysis

The AI Agent is the "brain" of the system, translating high-level, sometimes ambiguous, tasks posed by the user into a precise, ordered series of tool calls. Its implementation is independent of any specific model platform and built on a general, extensible architecture.

- Dynamic Task Planning:** The Agent's core capability lies in the `_plan_task` function. Rather than relying on hard-coded logic, it dynamically generates task plans by interacting with a larger model.
 - Self-Awareness and Prompt Construction:** At the beginning of planning, the Agent first examines all available tools and their descriptions. It then packages this tool information, the user's task, and strict rules (such as data transfer format) into a highly structured `planning_prompt`.
 - Model as Planner:** This prompt is fed into a general text-based model. The model reasoned based on the provided context and returned a multi-step action plan in JSON format. This design is highly scalable: as tools are added or modified in the system, the Agent's planning capabilities are automatically updated without requiring code modifications.
- Toolchain and Data Flow:** Real-world tasks often require the collaboration of multiple tools. For example, "take a picture and describe" requires the output (image path) of the "take a picture" tool to be used as the input of the "describe" tool. The AI Agent elegantly implements this through the `_process_step_parameters` function.
 - Data Reference Placeholders:** During the planning phase, large models embed special placeholders, such as `{{steps.1.outputs.data}}`, in parameter values where data needs to be passed.

- **Real-Time Parameter Replacement:** In the `_execute_agent_workflow` main loop, `_process_step_parameters` is called before each step. It uses regular expressions to scan all parameters of the current step. Upon discovering a placeholder, it finds the corresponding data from the output list of the previous step and replaces it in real time. This mechanism is key to automating complex tasks.
3. **Supervised Execution and Fault Tolerance:** `_execute_agent_workflow` constitutes the Agent's main execution loop. It strictly follows the planned sequence of steps, executing each action sequentially and ensuring data is correctly passed between them.
- **Atomic Steps:** Each step is treated as an independent "atomic operation." If any step fails, the entire task chain immediately aborts and reports an error. This ensures system stability and predictability, preventing continued execution in an erroneous state.

In summary, the general implementation of the AI Agent demonstrates an advanced software architecture: rather than directly solving a problem, it builds a framework that enables an external, general-purpose reasoning engine (a large model) to solve the problem. By leveraging two core mechanisms, dynamic programming and data flow management, the Agent orchestrates a series of independent tools into complex workflows capable of completing advanced tasks.

3. Practical Practice

3.1 Configuring Online LLM

Raspberry Pi 5 requires entering a Docker container; RDK X5 and Orin controllers do not:

```
./ros2_docker.sh
```

If you need to enter the same Docker container to input other commands later, simply type

```
./ros2_docker.sh
```

again in the host machine's terminal.

1. **Then you need to update the key in the configuration file. Open the model interface configuration file `large_model_interface.yaml`:**

```
vim ~/yahboom_ws/src/largemodel/config/large_model_interface.yaml
```

2. **Enter your API Key:**

Find the corresponding section and paste the API Key you just copied. Here's an example configuration using Tongyi Qianwen:

```
# large_model_interface.yaml

## Thousand Questions on Tongyi
qianwen_api_key: "sk-xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx" # Paste your Key
qianwen_model: "qwen-vl-max-latest" # You can choose the model as needed, such
as qwen-turbo, qwen-plus
```

3. **Open the main configuration file `yahboom.yaml`:**

```
vim ~/yahboom_ws/src/largemodel/config/yahboom.yaml
```

4. **Select the online platform to use:**

Modify the `llm_platform` parameter to the name of the platform you want to use

```
# yahboom.yaml

model_service:
  ros__parameters:
    # ...
    llm_platform: 'tongyi' #Optional platforms: 'ollama', 'tongyi', 'spark',
    'qianfan', 'openrouter'
```

After modifying the configuration file, you need to recompile and source it in the workspace:

```
cd ~/yahboom_ws
colcon build && source install/setup.bash
```

3.2 Starting and Testing the Functionality

1. Starting the `largemodel` main program:

Open a terminal, enter the Docker container, and then run the following command:

```
ros2 launch largemodel largemodel_control.launch.py
```

2. Testing:

- **Wake-up:** Say into the microphone, "Hello, yahboom."
- **Dialogue:** After the speaker responds, you can say: `Generate an image similar to the current scene based on the current environment`
- **Observe the logs:** In the terminal running the `launch` file, you should see:

The system receives a text command, invokes the `aiagent` tool, and then provides a prompt to the LLM (Linux Virtual Machine). The LLM analyzes the detailed steps of invoking the tool. For example, in this question, the `seewhat` tool is invoked to capture the image, which is then provided to the LLM for parsing. The parsed text is then fed back to the LLM as content for generating a new image.

- **Listen to the Answer:** Shortly, you should hear the answer from your speaker indicating completion, and a pop-up window will display the camera feed and the newly generated image. The newly generated image can be found later in the path `~/yahboom_ws/src/largemodel/resources_file/generated_images`.